Gondwana Research 134 (2024) 30–47

Contents lists available at ScienceDirect

Gondwana Research

journal homepage: www.elsevier.com/locate/gr

Paleomagnetic constraints on the link between the Comei-Bunbury large igneous province and the Kerguelen mantle plume

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ARTICLE INFO

Article history:
Received 19 February 2024
Revised 15 May 2024
Accepted 6 July 2024
Available online 7 July 2024
Handling Editor: J.G. Meert

Keywords:
Paleomagnetism
Tethyan Himalaya
Zhela Formation
Comei-Bunbury large igneous province
Kerguelen mantle plume

ABSTRACT

The causal link between the Comei-Bunbury large igneous province (CBLIP) and the Kerguelen mantle plume (KMP) remains a topic of debate. Here, a new paleomagnetic study was conducted on the Zhela Formation basalts and sandstones (ca. 147–140 Ma) in the Taga area of the eastern Tethyan Himalaya (TH). The sandstone specimens had erratic demagnetization patterns, and reliable characteristic remanent magnetization (ChRM) directions could not be isolated. Positive fold and reversal tests indicate that the ChRM directions of the Zhela Formation basalts should be the primary magnetization. The tilt-corrected site-mean direction from 18 sites is D/I = 87.8°/70.0° (α_{95} = 6.8°), yielded a paleopole of 21.6°N/130.2°E (A_{95} = 12.0°) and a paleolatitude of 54.7° ± 12.0°S for the study area (28.4°N, 91.8°E). Comparison with robust latest Jurassic and Early Cretaceous paleomagnetic, geochemical, and geochronological data from the eastern TH, northeastern Indian craton, and southwestern Australia reveal that (1) the eruption periods of the Comei igneous rocks are similar to the Bunbury basalt (and the Naturaliste Plateau volcanic rocks); (2) the original erupted positions of the Comei-Bunbury (CB) igneous rocks were in agreement with the reconstructed KMP; and (3) the CBLIP originated from the KMP.

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1. Introduction

Large igneous provinces (LIPs) are magmatic provinces that cover areas greater than 100,000 km^2, contain igneous volumes exceeding 100,000 km^3, and are usually characterized by an intense eruptive pulse(s) of short duration (~1–5 Myr) (Bryan and Ernst, 2008; Ernst et al., 2016). Because of their effect on regional and global climate, their influence on the coevolution of organisms and the environment, and their importance to tectonic reconstruction, the LIPs have been studied extensively in the Earth sciences (Coffin and Eldholm, 1994; Zhu et al., 2009, 2013; Ernst et al., 2013; Zhu et al., 2023).

The Kerguelen Plateau, renowned for its origin from the KMP, is one of the prominent LIPs on our planet (Larson, 1991; Frey et al., 1996; Zhu et al., 2008). The KMP and adjacent continental margin has led to the extensive scattering of igneous rocks from the Cretaceous and Cenozoic periods, due to the distinct movements experienced by the Indian, Australian, and Antarctic plates (Storey et al., 1992; Ingle et al., 2003; Gibbons et al., 2012; Chen et al., 2021). The rocks associated with the KMP predominantly consist of the Rajmahal (118–117 Ma; Klootwijk, 1971; Poornachandra Rao and Mallikharjuna Rao, 1996; Sherwood and Mallik, 1996), Ninetyeast Ridge (82–37 Ma; Frey and Weis, 1995; Coffin et al., 2002), Broken Ridge (95 Ma; Duncan, 2002; Coffin et al., 2002), and Southern (120–110 Ma; Mahoney et al., 1995; Duncan, 2002; Coffin et al., 2002), Central (105–100 Ma; Duncan, 2002; Coffin

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https://doi.org/10.1016/j.gr.2024.07.004
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et al., 2002), and Northern (40–24 Ma; Weis et al., 2002; Duncan, 2002; Coffin et al., 2002) Kerguelen Plateau (Fig. 1). However, there is no consensus regarding the connection between the latest Jurassic and Early Cretaceous igneous rocks (147–130 Ma) found around the circum-eastern Gondwana and the KMP. Some researchers have proposed a link between these igneous rocks (e.g., Comei igneous rocks in eastern TH, Bunbury basalt in southwestern Australia) and the KMP, suggesting that the plume played a pivotal role in the eastern Gondwana breakup (Ingle et al., 2002; Zhu et al., 2009; Shi et al., 2018; Chen et al., 2021; Peng et al., 2022). However, contrasting viewpoints exist regarding the formation of these latest Jurassic and Early Cretaceous igneous rocks. Other researchers have argued that the formation of these rocks could be attributed to decompression melting in the passive rifting process

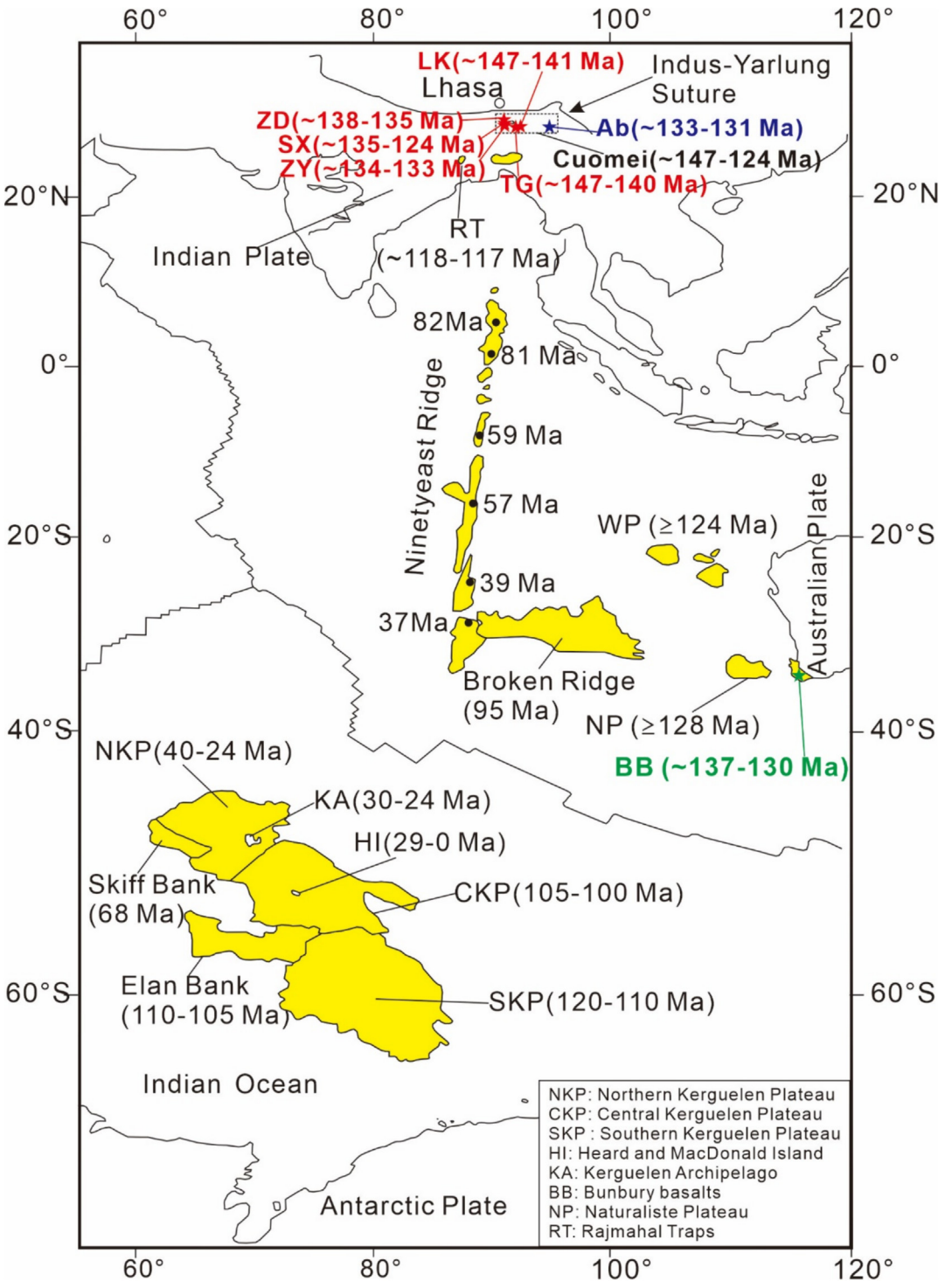


Fig. 1. Temporal and spatial locations of the CBLIP and products (yellow parts) in connection with the KMP (modified from Zhu et al., 2008). The red, blue, and green stars show sampling locations of previous latest Jurassic and Early Cretaceous paleomagnetic studies of the eastern TH, northeastern Indian craton, and southwestern Australia, respectively (abbreviation see Table 3). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

between the Indian Plate and the Australia-Antarctic Plate, suggesting that the eastern Gondwana breakup may not be directly linked to the KMP (Zeng et al., 2019; Olierook et al., 2016, 2019).

To better estimate the causal link between the latest Jurassic and Early Cretaceous igneous rocks and the KMP, it is vital to understand their spatial relationship throughout the latest Jurassic to Early Cretaceous. Based on the plate tectonic reconstruction, the KMP was positioned beneath the triple junction of the Australian, Antarctic, and Indian plates at ca. 135 Ma (Davies et al., 1989). However, other models predict that during the Early Cretaceous, the KMP was situated beneath Antarctica, leading to a significant distance (ca. 1000 km) between the KMP and the Bunbury basalt (Müller et al., 1993). Paleomagnetism is the most effective method to quantify plate paleogeography and plays a pivotal role in understanding the history of plate movement (Appel et al., 1998; Tan et al., 2010; Liebke et al., 2010; Yi et al., 2011; van Hinsbergen et al., 2012; Lippert et al., 2014; Ma et al., 2014, 2017, 2018; Yang et al., 2015a, 2019; Yan et al., 2016; Li et al., 2017; Cao et al., 2017; Tong et al., 2017, 2022; Song et al., 2020; Bian et al., 2021, 2022, 2023; Yuan et al., 2021, 2023; Jiao et al., 2023; Xu et al., 2024; Hou et al., 2024). Because these latest Jurassic and Early Cretaceous igneous rocks associated with the KMP have been scattered to various regions in the Australian, Antarctic, and Indian plates, and the TH was located at the northern margin of the Indian plate, reliable latest Jurassic and Early Cretaceous paleomagnetic data from the TH are essential for constraining their spatial relationship with the KMP. Considering that (1) some published latest Jurassic and Early Cretaceous paleomagnetic data from the TH lack robust field tests to question the reliability of the data (Klootwijk and Bingham, 1980; Zhu et al., 1981; Patzelt et al., 1996; Li et al., 2006; Huang et al., 2015); and (2) although these Jurassic and Cretaceous igneous rocks have an overall similarity in geochemical characteristics to the KMP (e.g., Zhu et al., 2007, 2008; Shi et al., 2018; Chen et al., 2021; Peng et al., 2022), the spatial relationship of these latest Jurassic and Early Cretaceous igneous rocks to the KMP is an essential prerequisite for understanding the causal link between the eastern Gondwana breakup and the KMP, additional reliable latest Jurassic and Early Cretaceous paleomagnetic datasets with well-established ages constraint from the TH are still required to address the aforementioned issues.

In this study, we present an analysis of paleomagnetic data on the Zhela Formation (Fm.) basalts and sandstones in the Taga area of the eastern TH. The new results allow us to constrain the paleogeography of the TH during the latest Jurassic to Early Cretaceous and thus provide a better understanding of the causal link between the CBLIP and the KMP.

2. Geological setting and paleomagnetic sampling

The TH is separated from the Greater Himalaya to the south by the South Tibetan Detachment System and from the Lhasa terrane to the north by the Indus-Yarlung Tsangpo suture zone (Fig. 2a). It mainly consists of sedimentary sequences ranging from Proterozoic to Eocene, with interbedded Paleozoic and Mesozoic volcanic rocks (Yin, 2006).

The area we are studying is located in the eastern TH where the Zhela Fm. volcano-sedimentary rocks are extensively exposed (Fig. 2b). The Zhela Fm., which is sandwiched unconformably between the overlying Weimei Fm. and the underlying Lure Fm., primarily include basalt, dacite, and siltstone. Zircon U-Pb dating indicate that the Zhela Fm. volcanic rocks in the Taga area formed at ca. 147–140 Ma (Deng, 2021; Liang et al., 2023). This is consistent with the bimodal magmatism (144–140 Ma) as observed in the Taga area by Zhang et al. (2023), as well as the youngest detrital zircons ages (ca. 138–127 Ma) documented by Jiao et al. (2021). The earliest folding of the Cretaceous strata in the TH predominantly occurred during the Eocene (Yin, 2006).

A total of 41 sites including 416 cores were collected from four sections during 2013, 2014, 2015, and 2023 (Fig. 2b). Thirty-two sites of them were collected from basalts (TG3–21, 24, 27, 33–36, BTG1–3) or sandstones (TG22, 23, 25, 26) distributed at two sections near Taga village (28°21′42.0″–28°22′14.0″N, 91°47′11.5″–91°47′34.5″E). Other 9 sites were collected from basalts (QZ1–9) distributed at two sections near Quzhemu village (28°16′58.0″–28°17′33.3″N, 91°45′56.9″–91°47′6.7″E). The bedding dips 28°–45° toward the northwest for the Taga sections and dips 19°–36° toward the southwest for the Quzhemu sections. The boundaries between two separate lava flows or interbedded sedimentary layers could be used to determine the bedding attitudes (Fig. 3).

Typically, each paleomagnetic site contained ten core samples and spanned several meters of the lateral outcrop. All core samples were drilled using a portable gasoline-powered and water-cooling drilling machine. Core samples were oriented using a magnetic compass and, when possible, a sun compass. The discrepancy in declination between these two readings was less than 2°, revealing that local magnetic correction can be neglected.

3. Methods

Oriented core samples with a 2.5 cm diameter were cut into standard 2.2 cm-long specimens in the laboratory. Approximately 91 % of the specimens were subjected to stepwise thermal demagnetization (TD) up to 580 °C or 690 °C using an ASC-TD 48 furnace. TD steps were set to 50–30 °C for low temperatures (<450 °C) and reduced to 30–5 °C for high temperatures (>450 °C). Alternatively, approximately 9 % of the specimens were treated with progressive alternating field demagnetization up to 80 mT or 140 mT using a D-2000 alternating field demagnetizer. Alternating field demagnetization steps were adjusted to 2.5–5 mT below 20 mT and increased to 10–20 mT above 20 mT. Specimens exhibiting relatively weak natural remanent magnetization (NRM) intensity were measured using a 2G-755-4k cryogenic magnetometer, whereas those with strong NRM intensity were measured using an AGICO JR-6A spinner magnetometer. The magnetometers and demagnetizers are placed in a magnetically shielded room in which the residual magnetic field is below 300 nT. Principal component analysis (Kirschvink, 1980) was used to determine the magnetic remanence directions of each specimen through at least four successive steps. Site-mean directions were then computed using Fisherian statistics (Fisher, 1953). The paleomagnetic software packages (Enkin, 1990; Cogné, 2003) were used to perform data analyses.

To determine minerals of magnetic carriers and evaluate the reliability of paleomagnetic data, the acquisition curves of isothermal remanent magnetization (IRM), backfield demagnetization of saturation IRM (SIRM), and TD of three-axis IRM (Lowrie, 1990) were conducted on five representative specimens. The major criterion used to choose specimens was the sampling sections (four sections) and the demagnetization results (demagnetized at ~580 °C or ~680 °C). IRMs were acquired using an ASC IM10-30 pulse magnetizer and measured using an AGICO JR-6A spinner magnetometer. The three-axis IRM was magnetized along three mutually orthogonal directions at the fields of 2.4 T, 0.4 T, and 0.12 T (Lowrie, 1990). The three-axis IRM underwent stepwise TD using a Magnetic Measurements Thermal Demagnetiser Super Cooling furnace. All rock magnetic and paleomagnetic experiments were performed at the Paleomagnetism and Environmental Magnetism Laboratory, China University of Geosciences, Beijing.

To better clarify the microtextures of the magnetic minerals of the Zhela Fm. basalts, and further elucidate whether the magnetic minerals were formed through exsolution at high temperature

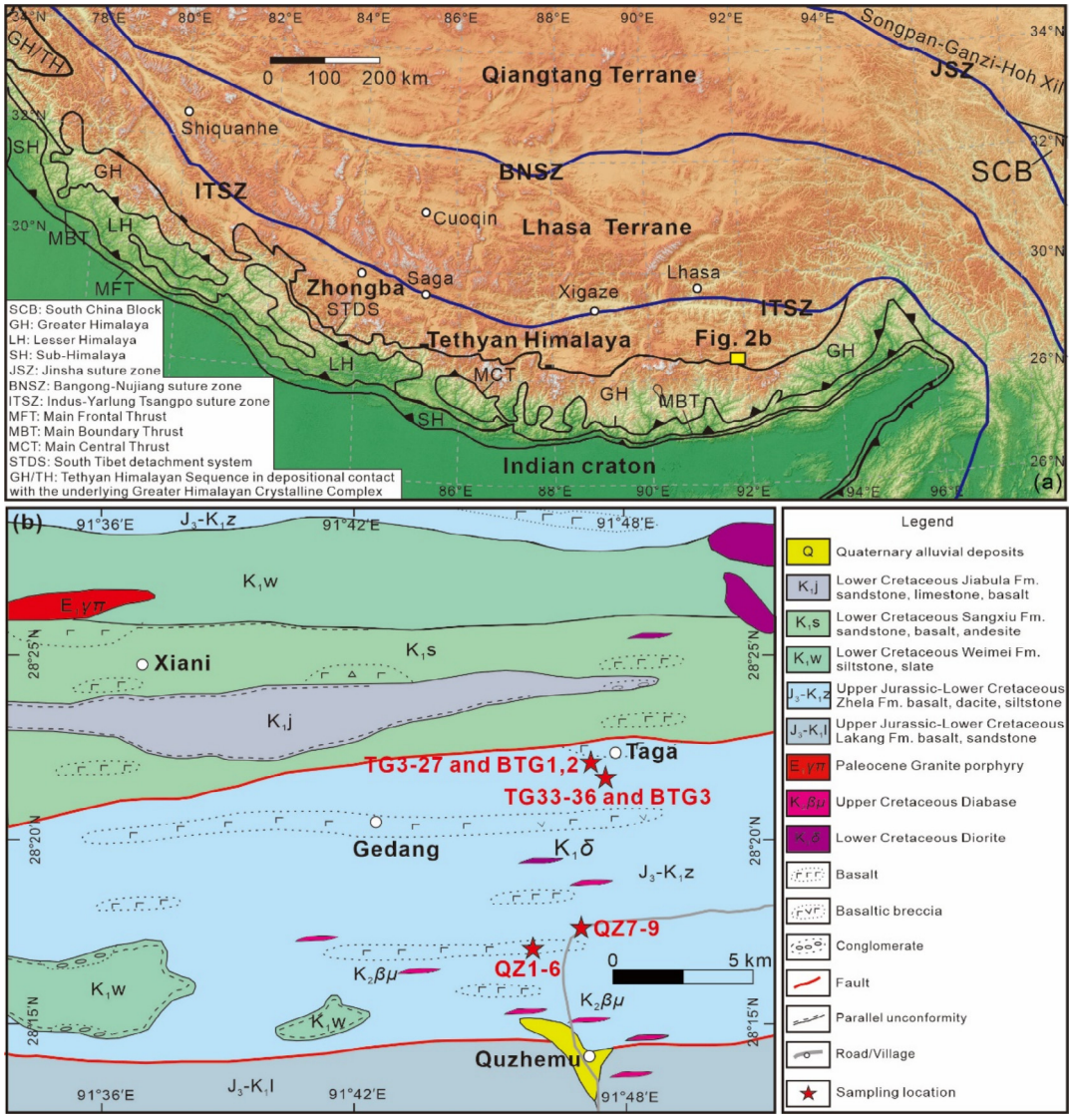


Fig. 2. (a) Schematic diagram of the Himalayan belt and surrounding areas (after Yang et al., 2015b). QGIS software (https://www.qgis.org/en/site/) and the Global Digital Elevation Model V003 data set from NASA Earth Data (http://search.earthdata.nasa.gov/search) were used to generate the map. (b) Geological map of the study area indicating paleomagnetic sampling sites.

during the initial cooling process, Scanning Electron Microscope (SEM) observation and Energy Dispersive Spectroscopy (EDS) analysis were performed on three selected samples using a Zeiss Merlin Compact SEM at Beijing SHRIMP Centre, Institute of Geology, Chinese Academy of Geological Sciences, Beijing. The Oxford X-MaxN 150 EDS detector was used to qualitatively determine the elemental composition of the magnetic carriers. The accelerating voltage is 15 kV and the working distance is ~8.6 mm.

4. Results

4.1. Rock magnetic results

Because no reliable ChRM directions could be isolated from the sandstone specimens (the following Section 4.3), only rock magnetic results from basalts are presented here. There are two distinct types within the rock magnetic results. For type 1, the IRM acquisition curves exhibit a relatively rapid increase below 100 mT, and saturation was not reached at 2.5 T (Fig. 4a, d, e). The backfield demagnetization of SIRM demonstrates a rapid decrease in remanence below 100 mT and a complete reduction to zero at 200–300 mT or ~50 mT. These findings reveal that these samples include both low- and high-coercivity magnetic carriers. Component analyses of coercivity distributions (Fig. 4f, i, j) display three-humped components: (1) high-coercivity component 1 (Fig. 4f, j) with $B_{1/2}$ (the field when half of the SIRM is acquired) of ~282–295 mT (contributing 82 %–87 % to the SIRM) or low-coercivity component 1 (Fig. 4i) with $B_{1/2}$ of ~32 mT (contributing 60 % to the SIRM); (2) low-coercivity component 2 (Fig. 4f, j) with $B_{1/2}$ of ~33–34 mT (contributing 7 %–10 % to the SIRM) or high-coercivity component 2 (Fig. 4i) with $B_{1/2}$ of ~468 mT (contributing 28 % to the SIRM); and (3) high-coercivity component 3 with $B_{1/2}$ of ~1585 mT (contributing 6 %–11 % to the SIRM). TD of three-axis IRM (Fig. 4f, i, j) reveals that the low-coercivity components unblocked at ~580 °C, suggesting that low-coercivity component is magnetite. The hard and medium components unblocked at ~680 °C, indicating that high-coercivity component is hematite. The high-coercivity component 3 with a minor contribution is interpreted to be used to fit the data (Fig. 4f, i) or the presence of goethite due to slight inflections at ~100–150 °C (Fig. 4o).

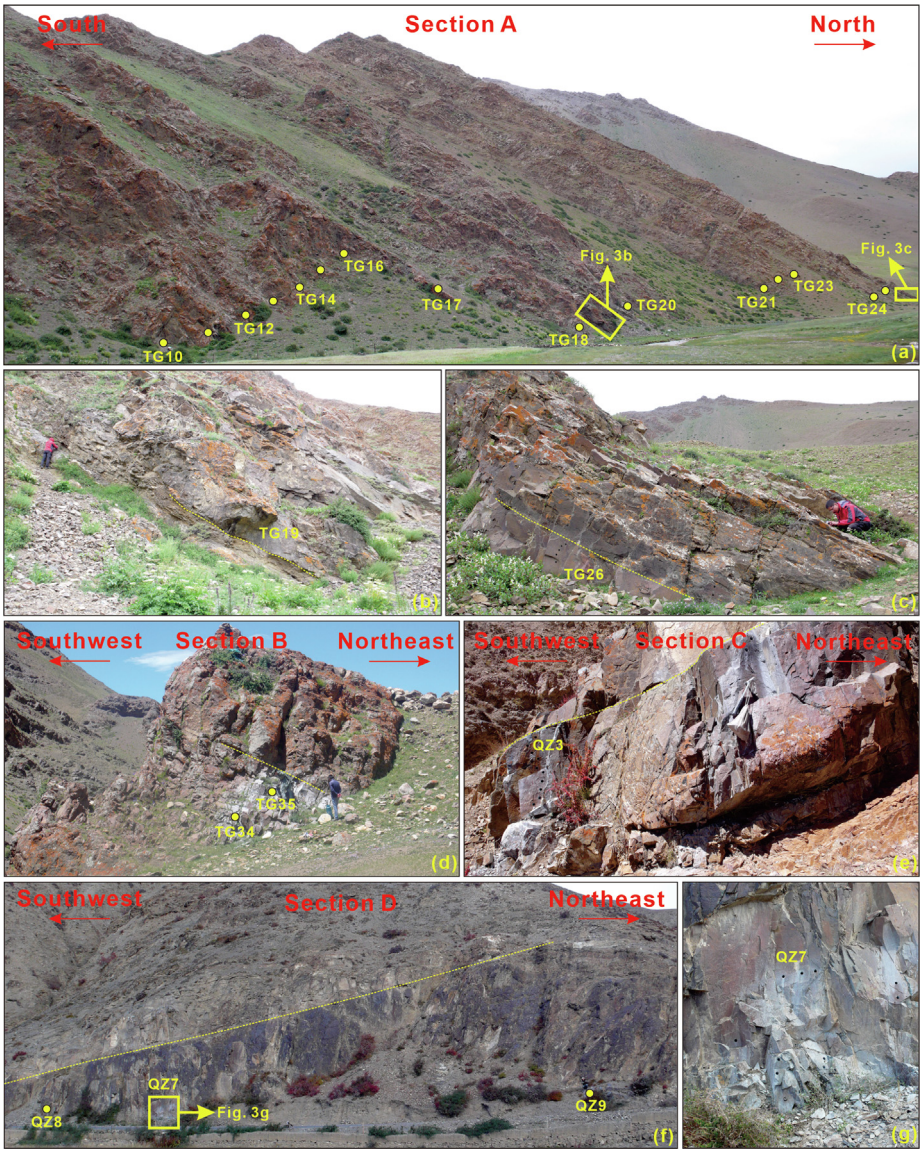


Fig. 3. (a–g) Field photographs of the sampled sections in the Taga area of the eastern TH.

For type 2, the IRM acquisition curves exhibit a rapid increase below 200 mT, and complete saturation is achieved at ~500 mT (Fig. 4b, c). The backfield demagnetization of SIRM shows a rapid decrease in remanence, reaching zero at ~50 mT. These findings reveal that low-coercivity magnetic carriers are dominant. Component analyses of coercivity distributions (Fig. 4g, h) demonstrate that only one component is present, namely the low-coercivity component with $B_{1/2}$ of ~35–49 mT (contributing 100 % to the SIRM). The three components unblocked at ~580 °C (Fig. 4l, m) further confirm that the low-coercivity magnetic carriers are magnetite.

4.2. Petrographic results

SEM observations indicate that some basalts have amygdaloidal structures (Fig. 5a) and that feldspar experienced no obvious deformation (Fig. 5a, d, g). EDS analyses indicate that the Zhela Fm. basalts are rich in Ti-Fe and Fe oxides (Fig. 5j–r). These oxide particles exhibit a lack of obvious oxidized rims and display various shapes, including rod-shaped and irregular shapes, ranging in sizes from less than 1 μm to several tens of microns. Our petrographic findings, in combination with the rock magnetic results, reveal that these Ti-Fe and Fe oxides were generated through exsolution at high temperature during the initial cooling process, rather than being the result of prolonged low-temperature alteration of the original Fe oxides.

4.3. Paleomagnetic results

Representative Zijderveld (1967) plots of the Zhela Fm. basalts and sandstones are shown in Fig. 6. The low-temperature (field) components from 233 specimens were isolated between room temperature and 300 °C (or 20 mT). Their mean direction (D/I = 3 55.8°/49.8°) in geographical coordinates (Fig. 7) shows a clustering around either the present local geomagnetic field direction (D/I = –0.1°/44.0°) or the geocentric axial dipole direction (D/I = 0.0°/47.2°), suggesting that it represents a recent overprint.

After the elimination of the low-temperature (field) components, a stable high-temperature (field) component that decayed towards the origin was resolved between 300 °C and 580 °C (or 690 °C) or between 20 mT and 80–140 mT. We define the high-temperature (field) component as ChRM. It is worth noting that

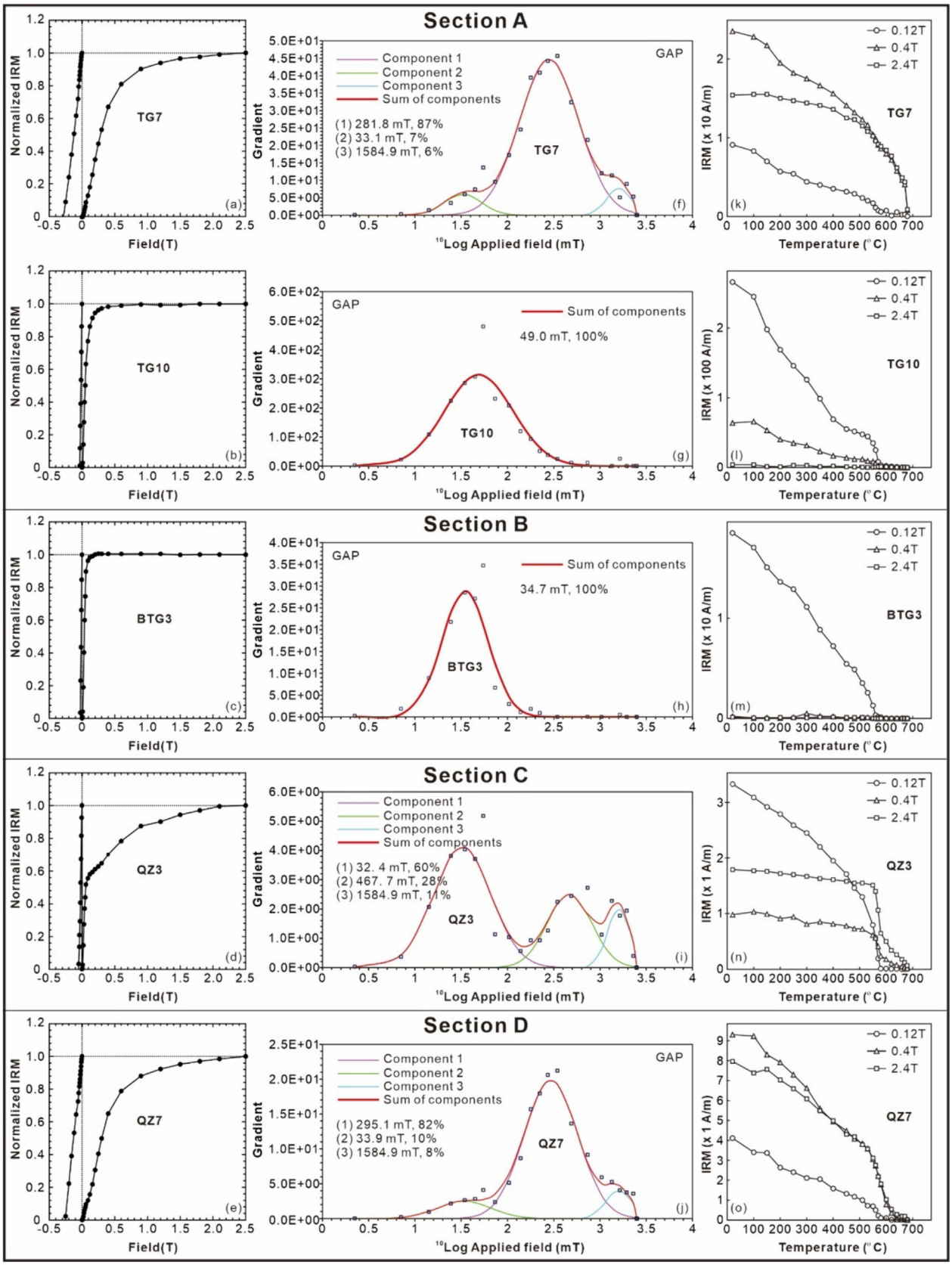


Fig. 4. (a–e) IRM acquisition curves and back-field demagnetization of the SIRM curves. (f–j) Results of IRM component analysis (Kruiver et al., 2001). (k–o) TD of the three-axis IRM curves. GAP indicates a gradient acquisition plot.

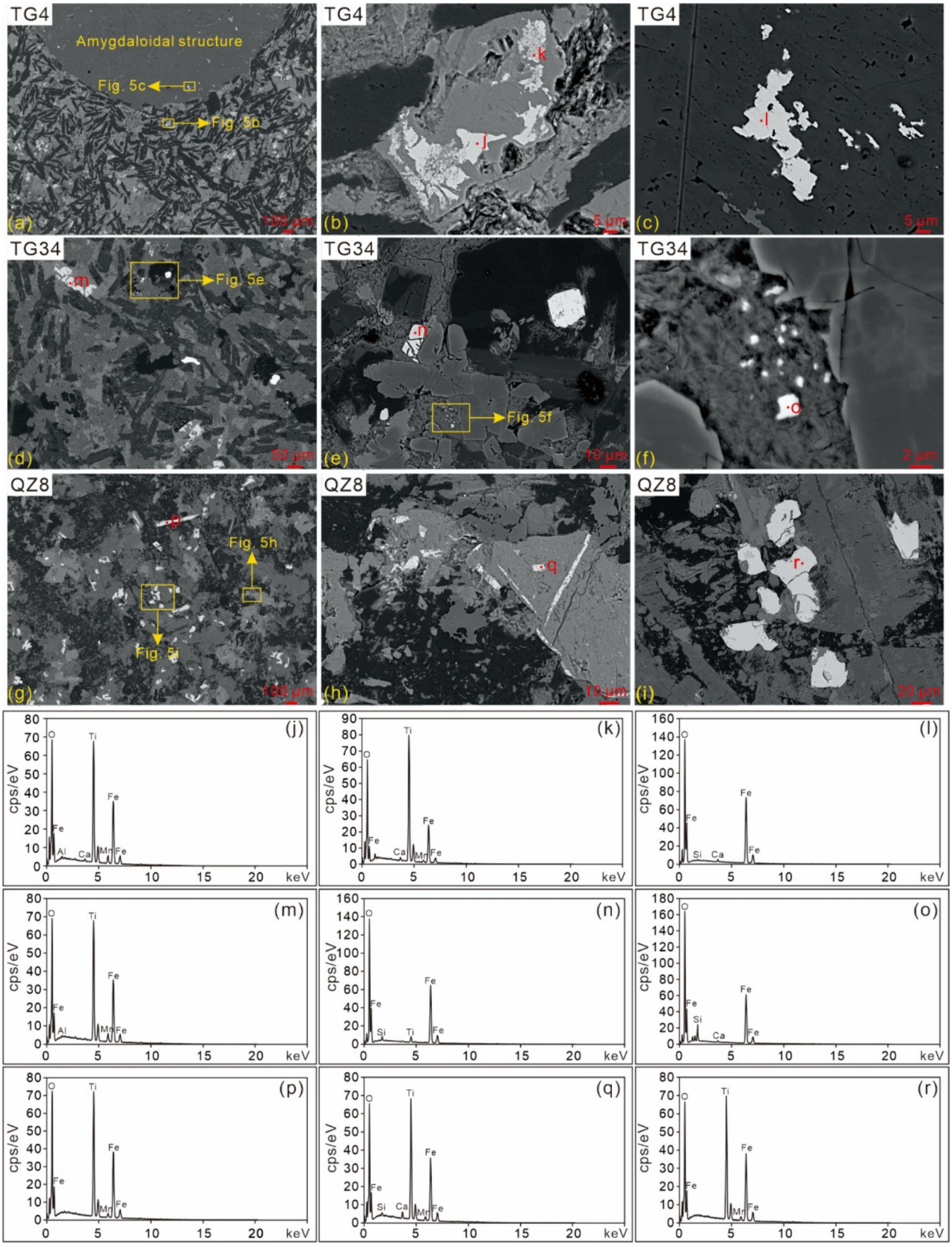


Fig. 5. (a–i) SEM observations. (j–r) EDS analyses. Red solid circles show the points analyzed by EDS. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

no reliable ChRM directions could be isolated from the sandstone specimens (Fig. 6s, t). Some specimens taken from slightly weathered basalts exhibited random demagnetization patterns (Fig. 6r), making it impossible to isolate stable ChRM directions. We

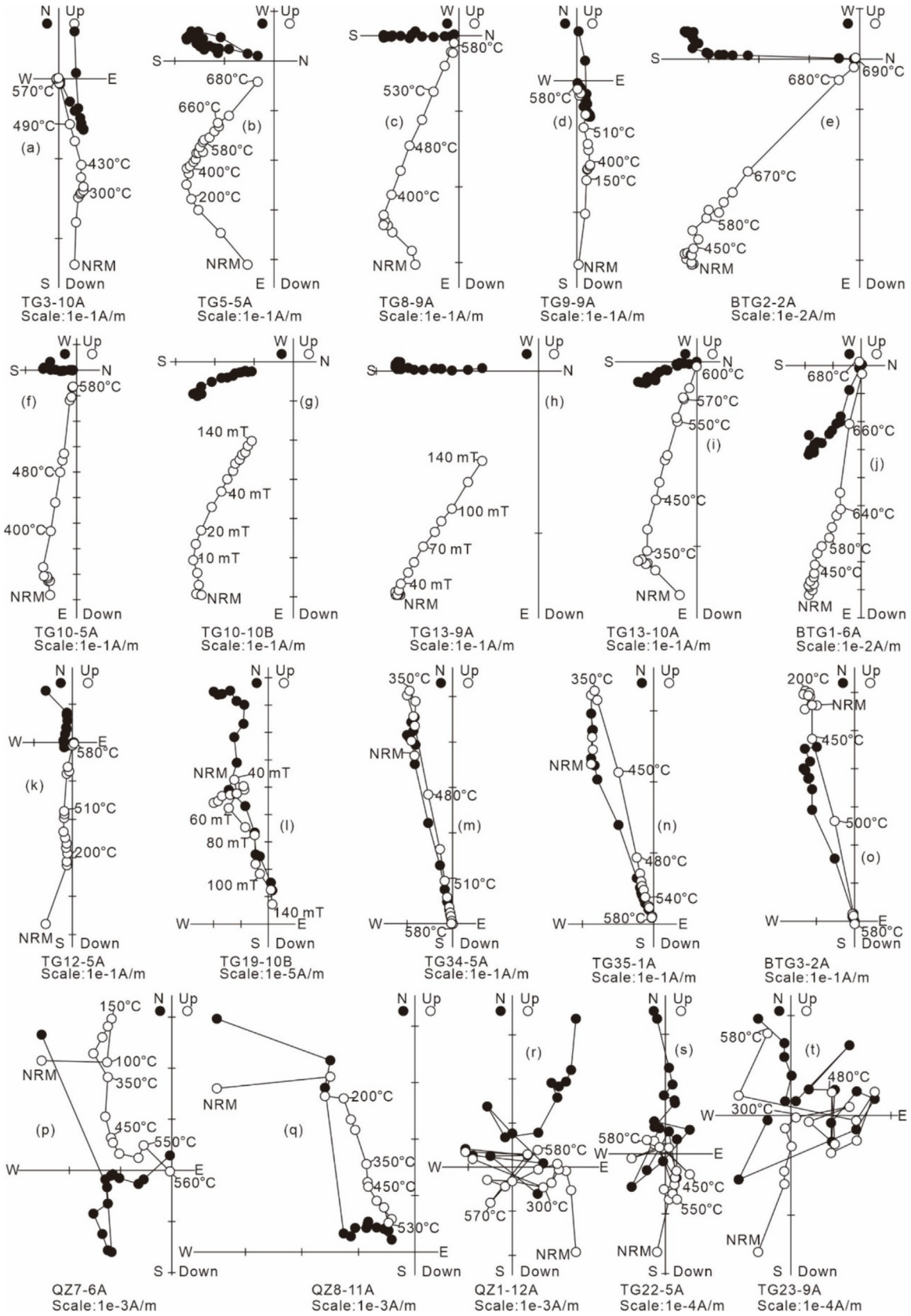


Fig. 6. Representative demagnetization diagrams for specimens from the Zhela Fm. basalts and sandstones in the Taga area of the eastern TH. Directions are plotted in geographical coordinates. The black (white) symbols represent projections onto the horizontal (vertical) plane.

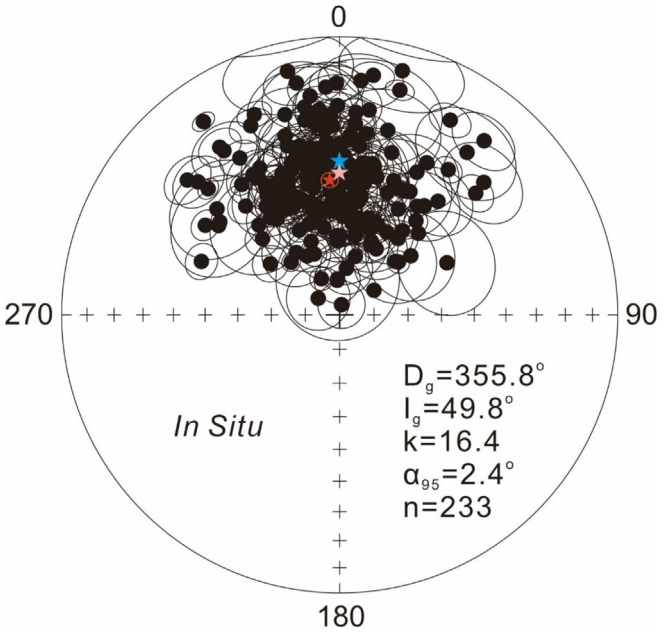


Fig. 7. Equal-area projection of the low-temperature (field) components. The red, blue, and pink stars show the mean directions of 233 specimens in geographical coordinates, the present local geomagnetic field direction, and the geocentric axial dipole direction, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

excluded sites based on the following filtering criteria: (1) sites with a mean direction calculated from specimens less than 5, and (2) sites with ChRM directions having α_{95} values greater than 15°, 22 out of 41 sites meet the criteria and provide the average direction (overall-mean A) of D_g/I_g = 161.9°/65.3° (k_g = 9.3, α_{95} = 10.8°) in situ, and D_s/I_s = 77.8°/68.2° (k_s = 14.3, α_{95} = 8.5°) after tilt-correction (Fig. 8a and Table 1). By averaging the virtual geomagnetic poles (N = 22), the site-mean paleopole is at 27.6°N, 133.7°E with A_{95} = 12.6° (Table 1). The site-mean directions pass McFadden (1990) fold tests at 95 % ($\xi_{in\ situ}$ = 10.48 > ξ_c = 5.46 > $\xi_{tilt\text{-}corrected}$ = 1.05) and 99 % ($\xi_{in\ situ}$ = 8.91 > ξ_c = 7.66 > $\xi_{tilt\text{-}corrected}$ = 2.74) confidence levels and yield a maximum grouping at 81.7 % (Fig. 8c; Watson and Enkin, 1993). The site-mean normal and reversed directions, however, fail the reversal test (McFadden and McElhinny, 1990). It is worth noting that the tilt-corrected ChRM directions from four sites (TG11, 12, 33, and QZ8) exhibit angular deviations of more than 28° from overall-mean A. These four sites may document a transitional direction (Vandamme, 1994). After excluding these four sites, the remaining 18 sites provide an average direction (overall-mean B) of D_g/I_g = 167.1°/60.2° (k_g = 10.5, α_{95} = 11.2°) in situ, and D_s/I_s = 87.8°/70.0° (k_s = 26.7, α_{95} = 6.8°) after tilt-correction (Fig. 8b and Table 1). By averaging the virtual geomagnetic poles (N = 18), the site-mean paleopole is at 21.6°N, 130.2°E with A_{95} = 12.0° (Table 1). The site-mean directions pass McFadden (1990) fold tests at a 95 % ($\xi_{in\ situ}$ = 6.28 > ξ_c = 4.94 > $\xi_{tilt\text{-}corrected}$ = 1.35) confidence level and yield a maximum grouping at 103 % (Fig. 8d; Watson and Enkin, 1993). Furthermore, positive results are also obtained from the reversal test of McFadden and McElhinny (1990). Based on these findings, along with the aforementioned rock magnetic and petrographic results, indicate that the ChRM directions obtained from the basalts of the Zhela Fm. are of primary origin.

We adopt the approach of Deenen et al. (2011, 2014) to evaluate whether our paleomagnetic findings from the basalts of the Zhela Fm. have averaged secular variation or not. The A_{95} value derived from our dataset is 11.0° (N = 18), which falls within the range of N-dependent A_{95min}/A_{95max} at 3.8°/13.3°. This result, in combination with the sampled Zhela Fm. basalts interbedded with multiple sedimentary layers, suggests that the site-mean direction of the Zhela Fm. basalts has averaged geomagnetic paleosecular variation.

5. Discussion

5.1. The geochronology of the Comei-Bunbury (CB) igneous rocks

The similarity in the emplacement age and the paleogeographic reconstruction of eastern Gondwana prompted Zhu et al. (2009) to link the Comei igneous rocks and Bunbury basalt (and the Naturaliste Plateau volcanic rocks) into a unified CBLIP. For the Comei igneous rocks, the ages vary from ca. 147 to ca. 124 Ma (e.g., Zhu et al., 2005, 2008, 2009; Liu et al., 2015; Wei et al., 2017; Shi et al., 2018; Zeng et al., 2019; Huang et al., 2019; Ding et al., 2020; Peng et al., 2022; Wang et al., 2023), with two pronounced peaks at ca. 141 Ma and ca. 133 Ma (Liang et al., 2023).

The Bunbury basalt was initially thought to have erupted at ca. 123 Ma and ca. 130 Ma (Frey et al., 1996). However, these ages are determined using $^{40}Ar/^{39}Ar$ dating of whole rocks. Because of the younging impact of alteration, the $^{40}Ar/^{39}Ar$ measurements on whole rocks have been shown to be unreliable (Verati and Jourdan, 2014). In contrast, $^{40}Ar/^{39}Ar$ studies of mineral grains (plagioclase) of Bunbury basalt yielded dates ranging from ca. 139 Ma to ca. 130 Ma (Fig. 9a; Table 2; Coffin et al., 2002; Olierook et al., 2016, 2019), with three clusters at ca. 137 Ma, ca. 133 Ma, and ca. 130 Ma (Olierook et al., 2016). This agrees with the kernel density estimate (KDE) from our simulation, which indicates that three possible peaks can be identified at ca. 137 Ma, ca. 132 Ma, and ca. 130 Ma (Fig. 9b). Notably, the last two clusters may have originated from the same flow based on Olierook et al. (2015).

The Naturaliste Plateau is located where two rift arms formed during the eastern Gondwana breakup (Direen et al., 2007). $^{40}Ar/^{39}Ar$ plagioclase ages and zircon U-Pb ages show that magmatic emplacement occurred between ca. 132 Ma and ca. 128 Ma (Table 2, Direen et al., 2017). Geochemical results demonstrate that the Naturaliste Plateau volcanic rocks and Bunbury basalt have comparable compositions (Direen et al., 2017). The ages of Naturaliste Plateau volcanic rocks and Bunbury basalt, therefore, have been combined. In the KDE plot, three possible peaks can be identified at ca. 137 Ma, ca. 132 Ma, and ca. 130 Ma (Fig. 9c), which is identical to the Bunbury basalt. The matching eruption periods of Comei igneous rocks and Bunbury basalt (and the Naturaliste Plateau volcanic rocks) suggest a potential common origin for the CB igneous rocks.

5.2. The paleolatitudes of the CBLIP

Understanding the causal link between the CBLIP and the KMP requires knowledge of the original position of emplacement of the CB igneous rocks. The latest Jurassic and Early Cretaceous paleomagnetic datasets from the CBLIP are presented in Table 3. To assess the reliability of these data, the updated *R* criteria (Meert et al., 2020) are applied. In this study, the paleomagnetic data must provide a robust field test (such as a positive fold test) or a positive reversal test (or dual-polarity ChRM directions), alongside fulfilling all other five *R* criteria. The paleolatitudes were determined based on the reference point located at their study area.

For the TH, 6 latest Jurassic to Early Cretaceous paleopoles (LK, TG, SX, ZD, ZY, and MC) met our selection criteria (Table 3). The remaining 6 paleopoles (TD, GB, JZ, MW, GC, and WL) were not considered for further discussion due to their lack of a reliable field test and/or insufficient number of sites or samples. Our new pale-

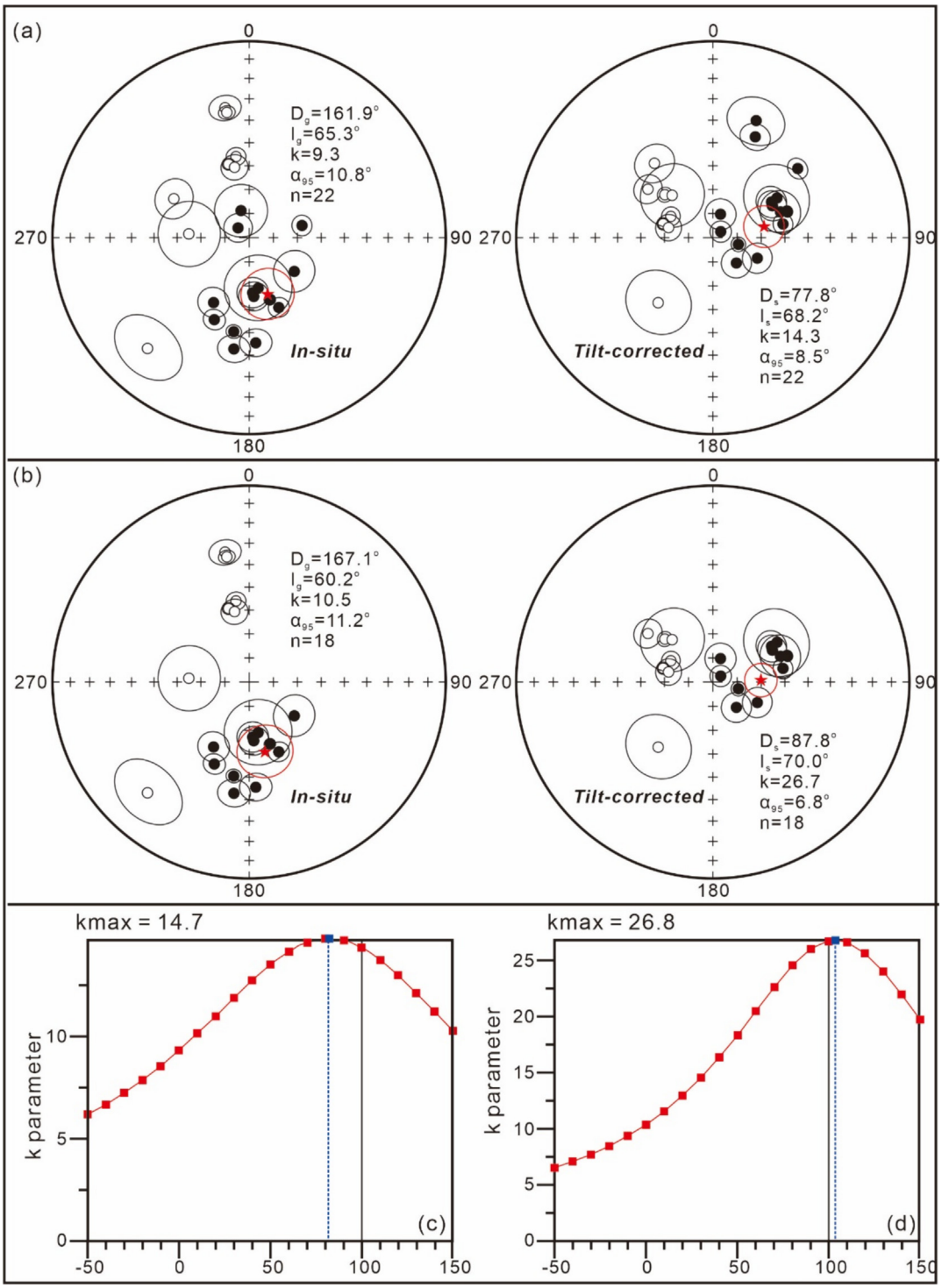


Fig. 8. (a, b) Equal-area projections of the site-mean ChRM directions from the Zhela Fm. basalts before and after tilt-correction. The stars show the overall mean directions of (a) 22 and (b) 18 sites. (c, d) Results of progressive unfolding of the site-mean direction from (c) 22 sites and (d) 18 sites.

omagnetic data, which meet all seven *R* criteria, provide a paleopole at 21.6°N/130.2°E (A_{95} = 12.0°) and reveal that the Taga area was located at 54.7° ± 12.0°S during ca. 147–140 Ma. The other five latest Jurassic and Early Cretaceous paleopoles were obtained from the Cuona, Langkazi, and Zhuode areas, respectively. For the Cuona area, Yang et al. (2015b) reported a paleopole of −36.8°N/315.2°E (A_{95} = 5.7°) from the Lakang Fm. lava flows, indicating that the Cuona area was positioned at 52.2° ± 5.7°S. The Lakang Fm. lava flows were assigned an Early Cretaceous (ca. 134–131 Ma, Yang et al., 2015b) based on the 1:250000 Longzi regional geological map (H46C004002, 2004). However, zircon U-Pb dating revealed that the Lakang Fm. lava flows were formed during ca. 147–141 Ma (Hou, 2017; Shi et al., 2018; Bian et al., 2019). These new geochronological findings have enabled us to determine the precise age of the Lakang Fm. lava flows. Furthermore, Meng et al. (2019) obtained a paleopole of 22.0°N/266.7°E (A_{95} = 7.6°) from the Lakang Fm. lava flows and limestones, revealing that the Cuona area was located at 39.6° ± 7.6°S during 137–125 Ma.

Table 1
Site-mean directions of the Zhela Fm. basalts and sandstones from the Taga area in the eastern TH.

Site	Strike/Dip (°)	n/N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	k	α_{95} (°)	Plat (°)	Plon (°)	Devi. (°)
TG3	293/44	6/7	157.0	58.3	79.0	60.2	253.2	4.2	26.1	147.2	8.0
TG4	293/44	8/8	203.0	52.6	52.5	85.9	158.4	4.4	33.1	99.5	18.2
TG5	296/41	7/10	189.3	49.9	104.7	79.1	399.1	3.0	21.3	113.7	13.0
TG6	296/41	No reliable site-mean direction									
TG7	296/41	8/10	176.4	45.4	114.8	69.7	79.7	6.2	9.4	125.0	13.2
TG8	301/45	10/11	175.5	65.5	61.8	61.9	63.5	6.1	38.9	147.6	9.2
TG9	301/45	8/9	161.5	62.8	70.6	57.0	559.1	2.3	31.4	153.0	11.7
TG10	301/45	7/10	169.9	68.7	58.2	58.5	20.0	13.8	41.3	153.0	13.0
[a]TG11	301/45	7/9	343.3	78.3	20.3	36.6	31.4	10.9	69.9	200.8	44.5
[a]TG12	301/45	9/10	311.4	83.9	22.8	43.6	77.0	5.9	69.5	184.7	37.4
TG13	301/45	7/9	176.3	67.1	58.6	61.4	89.9	6.4	41.2	148.6	10.6
TG14	301/45	No reliable site-mean direction									
TG15	301/45	9/11	349.5	−33.2	306.8	−56.1	77.1	5.9	8.0	312.2	25.0
TG16-18	291/40	No reliable site-mean direction									
TG19	291/40	11/12	349.9	−35.5	311.9	−63.2	213.9	3.1	4.8	303.9	22.0
TG20	291/40	No reliable site-mean direction									
TG21-27	297/45	No reliable site-mean direction									
BTG1	293/44	8/11	208.7	59.1	18.1	79.8	75.4	6.4	46.9	100.7	18.7
BTG2	296/41	6/8	187.8	42.3	137.1	75.7	118.9	6.2	7.5	110.0	18.8
[a]TG33	297/29	9/9	77.0	67.5	50.8	43.6	152.6	4.2	44.9	173.1	28.4
TG34	297/29	10/10	350.8	−55.6	296.9	−70.5	155.1	3.9	−9.1	303.3	13.7
TG35	297/29	11/11	344.4	−58.1	285.6	−68.6	337.0	2.5	−13.2	309.4	10.2
TG36	290/28	8/10	126.8	66.6	69.1	59.8	42.2	8.6	33.2	149.7	9.2
BTG3	297/29	7/8	348.3	−60.0	282.8	−71.2	127.1	5.4	−55.6	340.5	9.1
QZ1-2	137/30	No reliable site-mean direction									
QZ3	137/30	5/8	222.5	−24.5	220.0	−54.4	33.6	13.4	4.3	299.6	22.2
QZ4	137/30	No reliable site-mean direction									
QZ5-6	136/36	No reliable site-mean direction									
QZ7	117/19	6/13	273.6	−64.9	316.2	−65.6	26.8	13.2	20.4	306.2	22.2
[a]QZ8	117/19	10/11	297.4	−54.5	321.9	−50.2	36.4	8.1	−16.5	306.7	35.2
QZ9	117/19	No reliable site-mean direction									
Overall-mean A			161.9	65.3	77.8	68.2	14.3	8.5	27.6	133.7	
N = 22 sites									K = 7.1	A_{95} = 12.6	
Overall-mean B			167.1	60.2	87.8	70.0	26.7	6.8	21.6	130.2	
N = 18 sites									K = 9.3	A_{95} = 12.0	

Notes: Plat (Plon), latitude (longitude) of the pole; n/N, number of specimens used to calculate mean and measured; $D_g/I_g(D_s/I_s)$, declination and inclination in geographic (stratigraphic) coordinates; k (K), the best estimate of the precision parameter; α_{95}, 95 % confidence limit. Devi, the angular difference of site directions from the site mean direction of 22 sites. (1) Overall-mean A (N = 22): ①The McFadden (1990) fold test is positive at 95 % and 99 % confidence levels. "Xi" test: critical Xi at 95 % = 5.46 and 99 % = 7.66, respectively. "Xi1" and "Xi2" IS = 10.48 and 8.91, "Xi1" and "Xi2" TC = 1.05 and 2.74, respectively. (2) Overall-mean B (N = 18; without [a]sites): ①The McFadden (1990) fold test is positive at a 95 % confidence level. "Xi" test: critical Xi at 95 % = 4.94. "Xi1" IS = 6.28, "Xi1" TC = 1.35; ②The reversals test is positive at a 95 % confidence level. Normal polarity: D/I = 73.5°/70.0° (k = 33.6, n = 11); Reverse polarity: D/I = 288.6°/-68.0° (k = 25.1, n = 7); The angle between the two mean directions is $\gamma = 12.6° < \gamma_{critical} = 13.3°$; classification C.

For the Langkazi area, Ma et al. (2016) reported a paleopole of −5.9°N/308.0°E (A_{95} = 6.1°) from the Sangxiu Fm. lava flows, indicating that the Langkazi area was positioned at 48.5° ± 6.1°S during ca. 135–124 Ma. Additionally, Zhang et al. (2019) obtained a paleopole of 6.3°N/308.6°E (A_{95} = 9.1°) from the Sangxiu Fm. lava flows, revealing that the Langkazi area was located at 39.7° ± 9.1°S during 134–133 Ma. For the Zhuode area, Bian et al. (2019) reported a paleopole of 0.9°N/293.4°E (A_{95} = 7.0°) from the Zhela and Weimei Fms. volcanic rocks, indicating that the Zhuode area was positioned at 53.5° ± 7.0°S during ca. 138–135 Ma. Notably, the six reliable paleopoles (LK, TG, SX, ZD, ZY, and MC) obtained from the eastern Tethyan Himalaya are not distinguishable at the 95 % confidence interval because they align along a small circle with its center located at the reference point (28.5°N, 91.7°E) (Fig. 10). In summary, during the latest Jurassic to Early Cretaceous, the TH was positioned at a paleolatitude of ~54.7°S for the Taga area, ~52.2°–39.6°S for the Cuona area, ~48.5°–39.7°S for the Langkazi area, and ~53.5°S for the Zhuode area, respectively (Fig. 11).

For the northeastern Indian craton, Ali et al. (2012) obtained a paleopole of −24.6°N/313.6°E (dp/dm = 8.6°/10.2°) from the Abor volcanic rocks, indicating that the Siang area was located at 55.5° ± 8.6°S (Fig. 11). Ali et al. (2012) gave an Early Permian age to the Abor volcanic rocks. However, new zircon U-Pb dating reveals that the Abor volcanic rocks were formed during ca.133–131 Ma (Singh et al., 2019). For the southwestern Australian plate, Schmidt (1976) reported a paleopole of −50.0°N/163.0°E (A_{95} = 4.0°) from the Bunbury basalt, indicating that the Bunbury area was positioned at 52.0° ± 4.0°S (Fig. 11). According to Schmidt (1976), the Bunbury basalt was assigned a Late Cretaceous age (ca. 90 Ma) based on K/Ar dating conducted by McDougall and Wellman (1976). However, new $^{40}Ar/^{39}Ar$ dating indicates that the Bunbury basalt was formed during ca. 137–130 Ma (Olierook et al., 2016). These latest geochronological findings were used to determine the formation time of the Abor volcanic rocks and Bunbury basalt. In summary, the northeastern Indian craton and the southwestern Australian plate were positioned at ~55.5°S and ~52.0°S, respectively, during the Early Cretaceous.

5.3. The causal link between the CBLIP and the KMP

Based on (1) the distance between the Bunbury basalt and the proposed KMP is ~1,000 km (Müller et al., 1993; Frey et al., 1996); (2) the small volume of the CBLIP (~11,000 km^3) conflicts with the typical mantle plume models (Liu et al., 2015); (3) the eruption ages of Bunbury basalt (ca. 139–130 Ma) are older than those of the volcanism on the Kerguelen Plateau (ca. 120–

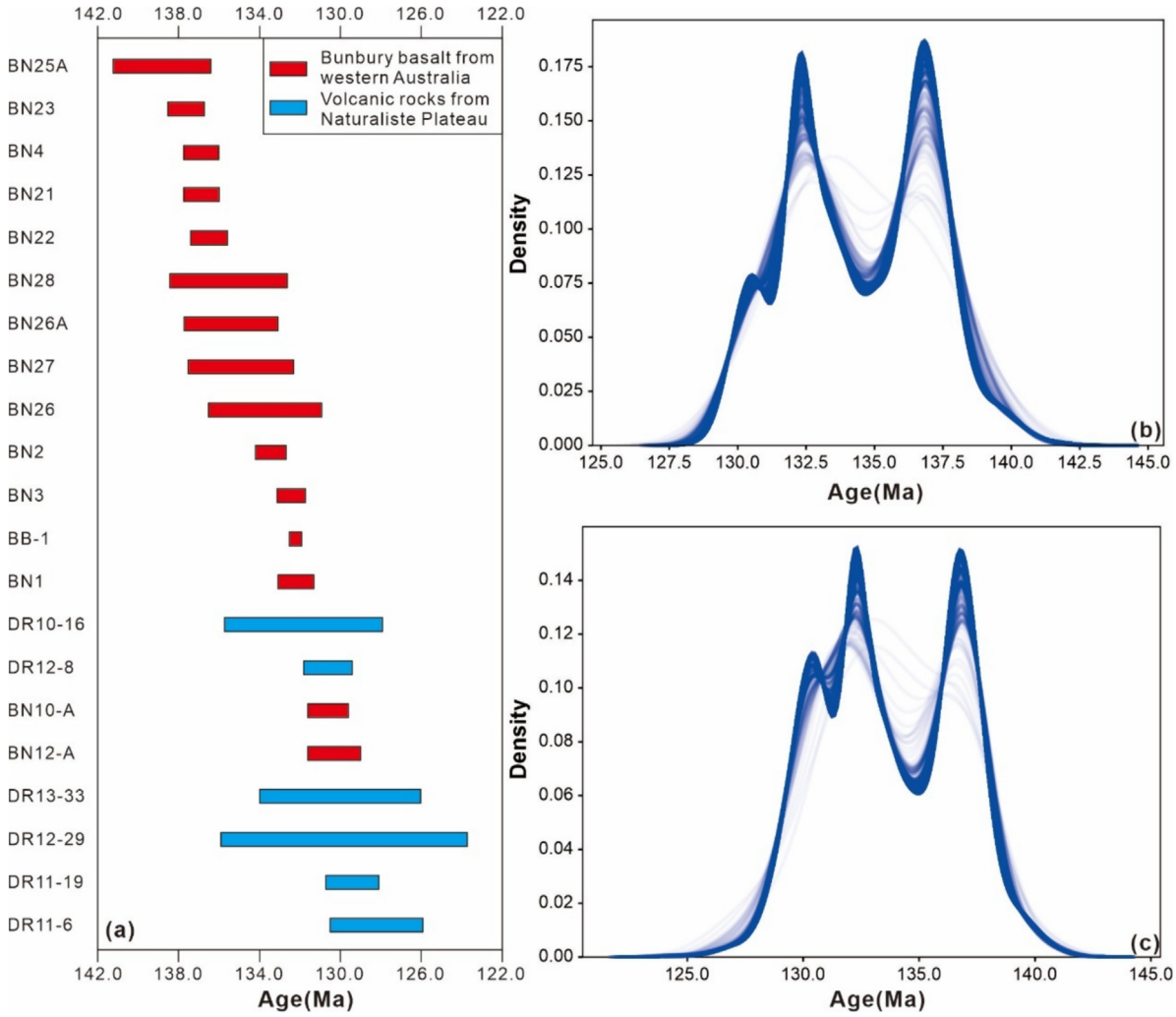


Fig. 9. (a) $^{40}Ar/^{39}Ar$ and zircon U–Pb ages for samples from the Bunbury basalt and the Naturaliste Plateau volcanic rocks. Please refer to Table 2 for the compilation of data and references. (b, c) Simulation of KDE variance using 1000 trials where dates are randomly selected for each of the ages in Table 2 (Vermeesch, 2012). The ages and uncertainties of the samples are used for this simulation, except for the data from Frey et al. (1996) which were determined using $^{40}Ar/^{39}Ar$ dating of whole rocks. Three potential peaks can be observed at ca. 137 Ma, ca. 132 Ma, and ca. 130 Ma in 999, 1000, and 787 trials (Bunbury basalt) for (b) and in 998, 987, and 926 trials (Bunbury basalt + Naturaliste Plateau volcanic rocks) for (c) in the KDE plot.

110 Ma); and (4) the Charong dolerites (~142 Ma) in the central Tethyan Himalaya are anticipated to melt at a lower temperature (1469 ± 44 °C) than a mantle plume (generally >1600 °C) (Zeng et al., 2019), some researchers have argued that these Comei igneous rocks and Bunbury basalt could be attributed to decompression melting-induced passive rifting process (Zeng et al., 2019; Olierook et al., 2016, 2019). Therefore, geochemical and spatial–temporal characteristics between the CBLIP and the KMP are essential for constraining their relationship.

In the TH, igneous rocks from the Latest Jurassic to Early Cretaceous are mostly found in the Zhela, Langkang, Weimei, and Sangxiu Fms. These igneous rocks had geochemical characteristics with alkali basalts from the KMP, including high TiO_2 contents, extensively fractionated light and heavy rare earth elements, and the absence of evident Eu anomalies (Zhu et al., 2008; Shi et al., 2018; Peng et al., 2022; Wang et al., 2023). Likewise, the Abor volcanic rocks and the Bunbury basalt (and the Naturaliste Plateau volcanic rocks) show similar geochemical characteristics with KMP products (Olierook et al., 2016; Direen et al., 2017; Singh et al., 2019). The geochemical results agree with the geochronological results, suggesting a potential common origin for the CB igneous rocks. Therefore, if the CB igneous rocks indeed came from the KMP, their original eruption position should be closer to the KMP. When the Indian and Australian plates separated, these igneous rocks became more scattered.

The original eruption position of the CB igneous rocks was located at a paleolatitude of 39.6–54.7°S, according to paleomagnetic findings. Their paleolatitudes are identical to the reconstructed KMP (~41.6–52.3°S; Torsvik et al., 2008; Fig. 12a). The spatial characteristics indicate a potential link between the CBLIP and the KMP. Therefore, the distance between the CBLIP and the proposed KMP does not provide compelling evidence against the hypothesis that the CB igneous rocks could be affected by the KMP.

Recently, the volume of the CB igneous rocks was recalculated by Liang et al. (2023). In addition to the Bunbury Basalt and the Comei igneous rocks, the volume of the CB igneous rocks also comprises Abor volcanic rocks in northeastern India, as well as volcanic rocks in southwestern Australia (mainly from Naturaliste Plateau, Wallaby Plateau, and Mentelle Bassin) based on their geochemical and geochronological similarities. The predicted volume

Table 2
$^{40}Ar/^{39}Ar$ and zircon U–Pb ages for samples from the Bunbury basalt and the Naturaliste Plateau volcanic rocks.

Sample	Rock type	Area	Slat (°N)	Slon (°E)	Age (Ma)	Technique	Reference
BB-3	Basalt	Western Australia	~33.3	115.6	130.6 ± 2.1	$^{40}Ar/^{39}Ar$	1
BB-4	Basalt	Western Australia	~33.3	115.6	129.2 ± 1.8	$^{40}Ar/^{39}Ar$	1
BB-93–1	Basalt	Western Australia	~33.3	115.6	123.7 ± 1.0	$^{40}Ar/^{39}Ar$	1
BB-93–2	Basalt	Western Australia	~33.3	115.6	123.1 ± 0.9	$^{40}Ar/^{39}Ar$	1
BB-1	Basalt	Western Australia	~33.3	115.6	132.2 ± 0.3	$^{40}Ar/^{39}Ar$	2
BN1	Basalt	Western Australia	−33.321	115.631	132.17 ± 0.89	$^{40}Ar/^{39}Ar$	3
BN2	Basalt	Western Australia	−33.317	115.633	133.43 ± 0.75	$^{40}Ar/^{39}Ar$	3
BN3	Basalt	Western Australia	−33.407	115.654	132.45 ± 0.65	$^{40}Ar/^{39}Ar$	3
BN4	Basalt	Western Australia	−33.407	115.654	136.90 ± 0.84	$^{40}Ar/^{39}Ar$	3
BN10-A	Basalt	Western Australia	−34.423	115.541	130.6 ± 1.0	$^{40}Ar/^{39}Ar$	3
BN12-A	Basalt	Western Australia	−34.422	115.551	130.3 ± 1.3	$^{40}Ar/^{39}Ar$	3
BN21	Basalt	Western Australia	−34.438	115.685	136.85 ± 0.88	$^{40}Ar/^{39}Ar$	3
BN22	Basalt	Western Australia	−34.481	115.689	136.50 ± 0.87	$^{40}Ar/^{39}Ar$	3
BN23	Basalt	Western Australia	−34.480	115.694	137.63 ± 0.89	$^{40}Ar/^{39}Ar$	3
BN25A	Basalt	Western Australia	−33.593	115.816	138.8 ± 2.4	$^{40}Ar/^{39}Ar$	4
BN26	Basalt	Western Australia	−33.593	115.816	133.7 ± 2.8	$^{40}Ar/^{39}Ar$	4
BN26A	Basalt	Western Australia	−33.593	115.816	135.4 ± 2.3	$^{40}Ar/^{39}Ar$	4
BN27	Basalt	Western Australia	−33.593	115.816	134.9 ± 2.6	$^{40}Ar/^{39}Ar$	4
BN28	Basalt	Western Australia	−33.773	115.812	135.5 ± 2.9	$^{40}Ar/^{39}Ar$	4
DR10-16	Granophyre	Naturaliste Plateau	−35.0	112.2	131.8 ± 3.9	U-Pb LA-ICP-MS	5
DR11-6	Monzodiorite	Naturaliste Plateau	−35.1	112.0	128.2 ± 2.3	U-Pb LA-ICP-MS	5
DR12-29	Rhyolite	Naturaliste Plateau	−35.1	111.7	129.8 ± 6.1	U-Pb LA-ICP-MS	5
DR11-19	Basalt	Naturaliste Plateau	−35.1	112.0	129.4 ± 1.3	$^{40}Ar/^{39}Ar$	5
DR12-8	Basalt	Naturaliste Plateau	−35.1	111.7	130.6 ± 1.2	$^{40}Ar/^{39}Ar$	5
DR13-33	Basalt	Naturaliste Plateau	−35.0	111.6	130.0 ± 4.0	$^{40}Ar/^{39}Ar$	5

Notes: Slat (Slon), latitude (Longitude) of samples. References: 1. Frey et al. (1996); 2. Coffin et al. (2002); 3. Olierook et al. (2016); 4. Olierook et al. (2019); 5. Direen et al. (2017).

Table 3
Summary of the latest Jurassic and Early Cretaceous paleopoles from the TH, northeastern Indian craton, and southwestern Australia.

ID	lithology	Area	Slat (°N)	Slon (°E)	Age (Ma)	Plat (°N)	Plon (°E)	A_{95}/(dp/dm) (dp/dm) (°)	Paleolat (°N)	n/N	Criterion (Q)	References
TH												
TD	sed	Dzong	28.8	83.8	early Aptian	12.0	289.0	6.0/7.5	−42.5 ± 6.0	95/–	123□5□7 (5)	1
GB	sed	Gamba	28.3	88.5	ca. 98–107	38.4	277.9	5.7/9.5	−22.7 ± 5.7	23/–	1□3□□□7 (3)	2
JZ	slate	Gyangze	28.9	89.8	J_3	7	329	11.0	−22.7 ± 11.0	11/–	1□3□5□7 (4)	3
MW	sed	Nielamu	28.7	86.3	J_3	−38.2	113.1	1.9/3.3	−18.6 ± 1.9	316/–	123□5□7 (5)	4
GC	sed	Nielamu	28.8	86.3	K_1	−32.7	115.7	5.1/8.4	−22.5 ± 5.1	56/–	123□5□7 (5)	4
LK	vol	Cuona	28.1	92.4	ca. 147–141	−26.8	315.2	5.7	−52.2 ± 5.7	225/31	123F5R7 (7)	5, 6, 7
TG	vol	Taga	28.4	91.8	ca. 147–140	21.6	130.2	12.0	−54.7 ± 12.0	142/18	123F5R7 (7)	This study
WL	vol	Wölong	28.5	87.0	ca. 138–130	4.4	256.0	4.6/5.2	−55.4 ± 4.6	201/–	123□5□7 (5)	8
SX	vol	Langkazi	28.8	91.3	ca. 135–124	−5.9	308.0	6.1	−48.5 ± 6.1	216/26	123F5R7 (7)	9
ZD	vol	Zhuode	28.9	91.3	ca. 138–135	0.9	293.4	7.0	−53.5 ± 7.0	219/31	123F5R7 (7)	10
ZY	vol	Langkazi	28.7	91.1	ca. 134–133	6.3	308.6	9.1	−39.7 ± 9.1	185/11	123F5□7 (6)	11
MC	vol + lim	Cuona	28.1	92.4	137–125	22.0	266.7	7.6	−39.6 ± 7.6	194/28	123F5R7 (7)	12
Northeastern Indian craton												
Ab	vol	Siang valley	28.3	95.1	ca. 133–131	−24.6	313.6	8.6/10.2	−55.5 ± 8.6	174/34	123□5D7 (6)	13, 14
Southwestern Australia												
BB	vol	Bunbury	−33.8	115.6	ca. 137–130	−50.0	163.0	4.0	−52.0 ± 4.0	54/5	123□5D7 (6)	15, 16

Notes: ID, paleopoles abbreviation used in the text; vol, volcanic rocks; sed, sedimentary rocks; lim, limestone; J_3, Late Jurassic; K_1, Early Cretaceous; Slat (Slon), latitude (longitude) of sites; Plat (Plon), latitude (longitude) of poles; A_{95}, 95 % confidence range; dp/dm, semi-axes of elliptical error of the pole at a probability of 95 %; Paleolat, paleolatitude calculated for the reference point located at their studied area; n/N, number of samples or sites used to calculate Fisher mean; Criteria (R), data quality criteria (number of criteria met) after Meert et al. (2020) (1, well determined rock age and a presumption that magnetization is the same age; 2, techniques and statistical analysis; 3, evaluation of remanence carriers; 4, field tests that constrain age of magnetization; 5, structural control and tectonic coherence with craton or block involved; inclination shallowing assessed in clastic sedimentary rocks; 6, the presence of reversals; 7, no resemblance to paleopoles of younger ages [by more than a period]; F, positive fold test; R, positive reversal test; D, dual-polarity; "□", failed to meet this criterion). References: 1. Klootwijk & Bingham (1980); 2. Patzelt et al. (1996); 3. Zhu et al. (1981); 4. Li et al. (2006); 5. Yang et al. (2015b); 6. Shi et al. (2018); 7. Hou (2017); 8. Huang et al. (2015); 9. Ma et al. (2016); 10. Bian et al. (2019); 11. Zhang et al. (2019); 12. Meng et al. (2019); 13. Ali et al. (2012); 14. Singh et al. (2019); 15. Schmidt (1976); 16. Olierook et al. (2016).

(114,250 km^3; Liang et al., 2023) corresponds well with the LIPs (Bryan and Ernst, 2008). Therefore, the volume of the CBLIP is consistent with the typical mantle plume models.

The previous studies proposed that the initial massive pulse of the KMP occurred around 120–110 Ma (Coffin et al., 2002). However, the CB igneous rocks whose crystallization ages ranged from

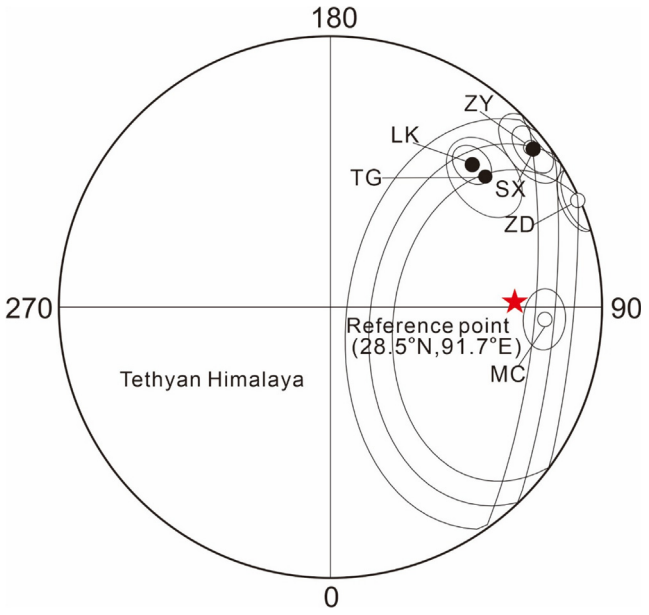


Fig. 10. Reliable latest Jurassic and Early Cretaceous paleopoles obtained from the Tethyan Himalaya. The small circle passing through the six paleopoles is centered on the reference point 28.5°N, 91.7°E.

147 Ma to 125 Ma were thought to have similar major, trace elements, and Sr-Nd-Pb isotopic compositions to KMP (Zhu et al., 2008; Shi et al., 2018; Chen et al., 2021; Peng et al., 2022; Wang et al., 2023). Therefore, the eruption ages of CB igneous rocks older than those of the volcanism on the Kerguelen Plateau can be attributed to the latest Jurassic or the earliest Cretaceous incubation of the KMP (Fig. 12b; Zhu et al., 2008; Shi et al., 2018; Wang et al., 2023).

Although the Charong dolerites (~142 Ma) in the central Tethyan Himalaya formed at a lower temperature than a mantle plume (Zeng et al., 2019), the picritic porphyrites and Cuona basalts in the eastern Tethyan Himalaya originated at a high mantle potential temperature of >1550 °C (Xia et al., 2014) and 1502–1564 °C (Chen et al., 2021), respectively. Further studies to estimate the environment in which magma developed are still necessary for future work. Overall, the geochemical results, together with the spatial–temporal characteristics, support the interpretation that the CBLIP originated from the KMP.

6. Conclusions

We present a new paleomagnetic result from the Zhela Fm. basalts in the Taga area of the eastern TH. Our results, along with geochronological, geochemical, and paleomagnetic data from the CBLIP, allow us to draw the following conclusions:

(1) The Tage area of the eastern TH was located at 54.7° ± 12.0°S at ca. 147–140 Ma.
(2) The eruption periods of the Comei igneous rocks are consistent with the Bunbury basalt (and the Naturaliste Plateau volcanic rocks).
(3) The paleolatitudes of the CBLIP are identical to the reconstructed KMP during the latest Jurassic and Early Cretaceous (~147–124 Ma).
(4) The CBLIP originated from the KMP based on their similar geochemical and spatial–temporal characteristics.

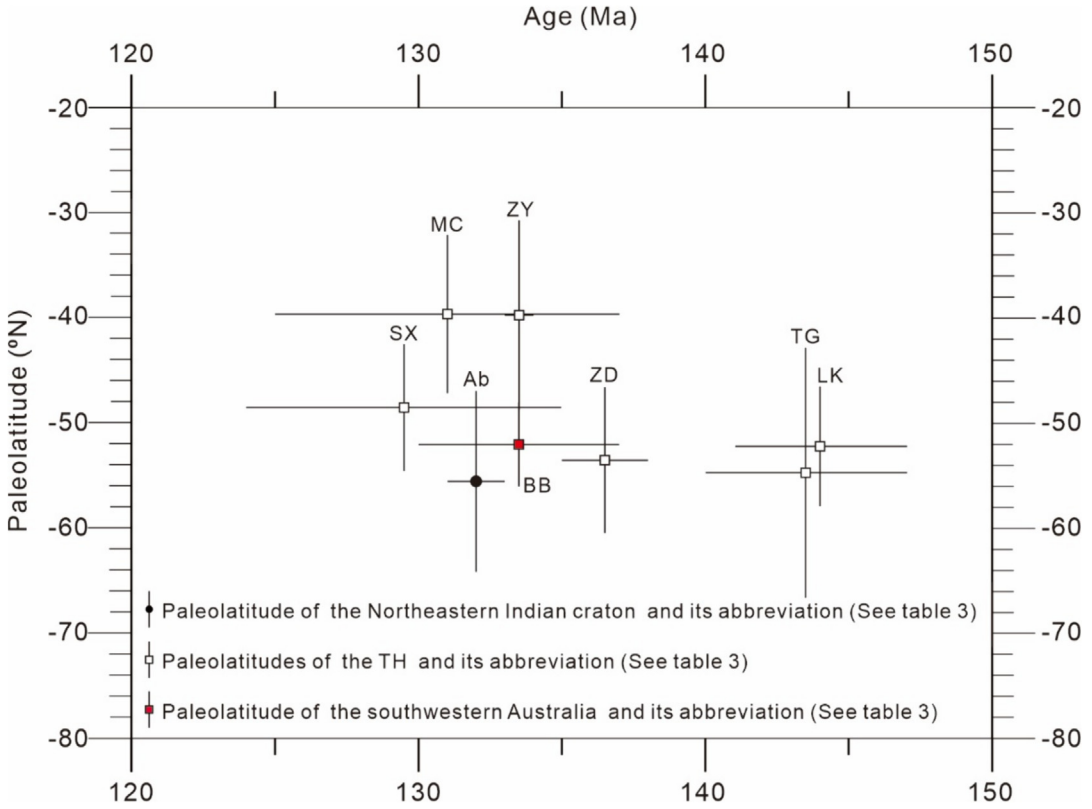


Fig. 11. Paleolatitude plots for the TH, the northeastern Indian craton, and the southwestern Australia. Paleolatitudes are calculated for the reference point located at their studied area.

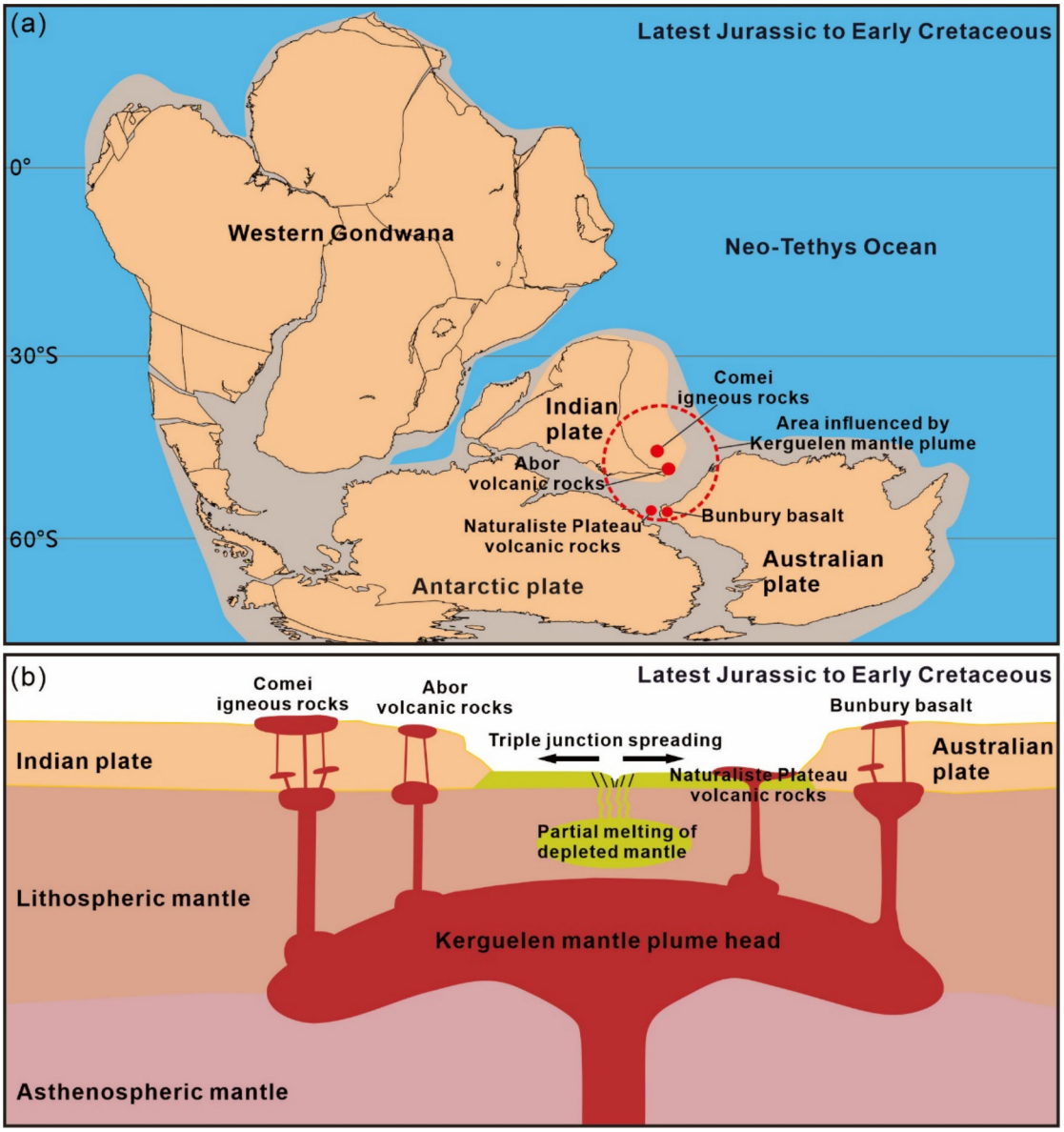


Fig. 12. (a) Reconstruction of Gondwana at the latest Jurassic to Early Cretaceous (after Zhu et al., 2022). (b) A schematic model illustrating the relationship between CBLIP and the KMP activity (after Peng et al., 2022).

CRediT authorship contribution statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We sincerely thank Editors M. Warrier Santosh, Joseph G. Meert, and two anonymous reviewers for their insightful comments and suggestions. This research was supported by the National Natural Science Foundation of China (42004050, 42072257, 41630104, and 41572205), the Fundamental Research Funds for the Central Universities (2652022001), the Frontiers Science Center for Deep-time Digital Earth (2652023001), and the Chinese "111" Project (B20011). We thank Siqi Wang and Qian Wen for laboratory assistance.

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